Trastuzumab rezetecan, a HER2-directed antibody–drug conjugate, in patients with advanced *HER2*-mutant non-small-cell lung cancer (HORIZON-Lung): phase 2 results from a multicentre, single-arm study

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Summary

Background Trastuzumab rezetecan (also known as SHR-A1811) is a novel antibody–drug conjugate consisting of a humanised HER2-directed monoclonal antibody, cleavable tetrapeptide-based linker, and DNA topoisomerase I inhibitor. In the phase 1 portion of this phase 1/2 study, trastuzumab rezetecan showed preliminary anti-tumour activity and a favourable safety profile in patients with *HER2*-mutant non-small-cell lung cancer (NSCLC). We present phase 2 results from the study, which aimed to further evaluate the activity and safety of trastuzumab rezetecan at the recommended dose.

Methods In this multicentre, single-arm, phase 2 trial, conducted in 35 hospitals in China, we recruited patients aged 18–75 years, with locally advanced or metastatic NSCLC with an activating *HER2* mutation and an Eastern Cooperative Oncology Group performance status score of 0–1, who had disease progression after or were intolerant to platinum-based chemotherapy and anti-PD-1 treatment or anti-PD-L1 treatment. Trastuzumab rezetecan was administered at 4·8 mg/kg intravenously once every 3 weeks. The primary endpoint was objective response rate assessed by an independent review committee in patients who received at least one cycle of study treatment. All patients who received at least one cycle of study treatment were included in safety analyses. This study is registered with ClinicalTrials.gov, NCT04818333, and is ongoing but not recruiting.

Findings Between April 14, 2023, and Dec 14, 2023, 94 patients were enrolled and treated. 42 (45%) patients were male, 52 (55%) female, 92 (98%) were Han Chinese, and two (2%) were other ethnicity Chinese. At data cutoff (June 14, 2024), the median duration of follow-up was 8·7 months (IQR 7·0–10·4). 69 (73%; 95% CI 63·3–82·0) of 94 patients had a confirmed objective response, as assessed by independent review committee. The most common grade 3–4 treatment-related adverse events were decreased neutrophil count (38 [40%] patients), decreased white blood cell count (25 [27%]), anaemia (22 [23%]), decreased platelet count (10 [11%]), and decreased lymphocyte count (seven [7%]). Treatment-related serious adverse events occurred in 22 (23%) patients, which were decreased platelet count (six [6%]), decreased neutrophil count (six [6%]), interstitial lung disease (five [5%]), decreased white blood cell count (four [4%]), anaemia (four [4%]), vomiting (three [3%]), pneumonia (three [3%]), hyponatraemia (two [2%]), and pyrexia (one [1%]), small intestinal obstruction (one [1%]), nausea (one [1%]), and chronic obstructive pulmonary disease (one [1%]). There were no treatment-related deaths.

Interpretation Trastuzumab rezetecan showed clinically meaningful activity and manageable safety in patients with previously treated *HER2*-mutant NSCLC. Further trials are justified.

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For the Chinese translation of the abstract see Online for appendix 1

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Introduction

HER2 is a tyrosine kinase receptor from the HER family that is involved in cell proliferation, differentiation, and survival.[1] Activating *HER2* (also known as *ERBB2*) mutations are found in approximately 2–4% of non-small-cell lung cancers (NSCLC), predominantly in adenocarcinomas.[1,2] Despite numerous attempts to target *HER2*-mutant tumours, monoclonal antibodies have failed to show clinical benefits,[3] whereas pan-HER tyrosine kinase inhibitors have mostly shown modest activity, with reported objective response rates of up to 30% and median progression-free survival of up to 6·9 months.[4–9] In 2024, zongertinib (BI 1810631; NCT04886804) and BAY 2927088 (NCT05099172)

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Research in context

Evidence before this study

We searched PubMed for clinical trials published in English between Jan 1, 2014, and Sep 30, 2024, using the terms "HER2-targeted" OR "*HER2*-mutant" OR "ERBB2-targeted" OR "*ERBB2*-mutant" AND "non-small cell lung cancer" OR "NSCLC", as well as abstract records of the American Society of Clinical Oncology, European Society of Medical Oncology, American Association for Cancer Research, and World Conference on Lung Cancer, using the same search terms. We identified a number of phase 1 and phase 2 trials that evaluated single agent anti-HER2 therapies, including tyrosine kinase inhibitors (dacomitinib, neratinib, afatinib, pyrotinib, poziotinib, BAY 2927088, and zongertinib) and antibody–drug conjugates (trastuzumab emtansine, trastuzumab deruxtecan, and trastuzumab rezetecan [SHR-A1811]), in pretreated patients with *HER2*-mutant non-small-cell lung cancer (NSCLC). The various HER2-targeted tyrosine kinase inhibitors have mostly shown modest activity, with data from 2024 for BAY 2927088 and zongertinib showing promise. Trastuzumab emtansine (drug–antibody ratio 3·1) showed clinical activity, but a relatively short duration of response. Trastuzumab deruxtecan (drug–antibody ratio 8) showed durable clinical activity and is the first approved HER2-directed therapy in NSCLC. Trastuzumab rezetecan is a novel antibody–drug conjugate (drug–antibody ratio 6) with a highly permeable and potent cytotoxic drug payload. Trastuzumab rezetecan showed preliminary anti-tumour activity and a favourable safety profile in patients with *HER2*-mutant NSCLC in the phase 1 portion of this phase 1/2 study.

Added value of this study

In this phase 2 portion, trastuzumab rezetecan at the recommended dose showed substantial and durable activity in pretreated patients with *HER2*-mutant NSCLC, leading to a clinically meaningful progression-free survival benefit. To the best of our knowledge, the objective response rate achieved with trastuzumab rezetecan is the highest reported among registrational studies for *HER2*-mutant NSCLC. The activity with trastuzumab rezetecan was consistent across a range of patients, regardless of previous lines of therapy, previous anti-HER2 tyrosine kinase inhibitor treatment, or the presence of brain metastases at baseline. Exploratory biomarker analysis further supported activity across various *HER2* mutations, including single-nucleotide variants and exon 20 insertions. The safety profile was manageable, with a low incidence of treatment discontinuation due to treatment-related adverse events.

Implications of all the available evidence

The combined findings from this phase 1/2 study support a favourable benefit-to-risk profile of trastuzumab rezetecan in pretreated patients with NSCLC harbouring an activating *HER2* mutation who progressed after previous systemic therapy, offering a potential new treatment option for this population. Future research should involve a randomised, controlled trial to assess trastuzumab rezetecan compared with chemotherapy plus immunotherapy in first-line treatment for advanced *HER2*-mutant NSCLC.

emerged as new agents showing encouraging activity.[10,11] Currently, the first-line standard-of-care treatment for *HER2*-mutant NSCLC involves platinum-based chemotherapy and immunotherapy. There are few later-line treatment options,[12] highlighting a crucial unmet need for more effective therapies for patients with late-stage disease.

Antibody–drug conjugates are a new class of therapeutic agent that combines a cytotoxic drug (ie, a payload) with a tumour-targeting antibody for precise delivery of toxins. Trastuzumab emtansine is the first antibody–drug conjugate to show activity in *HER2*-mutant NSCLC, with an objective response rate of 44% (95% CI 22–69), but the duration of response of 4 months (range 2–9) was relatively short.[13] Trastuzumab deruxtecan is a new-generation HER2-directed antibody–drug conjugate characterised by a drug–antibody ratio (ie, deruxtecan to trastuzumab) of 8 and membrane-permeable drugs, allowing for direct cytotoxicity in target tumour cells and a bystander anti-tumour effect on neighbouring cells, regardless of HER2 expression.[14,15] Trastuzumab deruxtecan has been approved by both the US Food and Drug Administration and the European Medicines Agency, and is approved in several countries for patients with previously treated *HER2*-mutant NSCLC, based on the single-arm global DESTINY-Lung01 and 02 trials.[16–18] At the approved dose of 5·4 mg/kg, trastuzumab deruxtecan has an objective response rate of 50% (95% CI 39·9–60·1), a median duration of response of 12·6 months (6·4–not reached), and a median progression-free survival of 10·0 months (7·7–15·2).[18]

Trastuzumab rezetecan (also known as SHR-A1811; Jiangsu Hengrui Pharmaceuticals, Suzhou, China) is a novel antibody–drug conjugate, composed of a humanised HER2-directed monoclonal antibody conjugated to a DNA topoisomerase I inhibitor drug (SHR169265) via a tetrapeptide-based cleavable linker.[19] The payload shows high membrane permeability and potent cytotoxicity, enabling strong on-target and bystander killing effects.[19] To optimise its benefit–risk profile, trastuzumab rezetecan is designed with a drug–antibody ratio of 6 and features a chiral cyclopropyl group between the linker and toxin to enhance stability and prevent early toxin release in circulation. In its first-in-human study, trastuzumab rezetecan showed preliminary activity with favourable safety across various advanced solid tumours at doses selected for expansion.[20] In the phase 1 portion of this ongoing phase 1/2 trial in advanced *HER2*-mutant NSCLC, trastuzumab rezetecan showed durable anti-cancer activity in patients with

HER2-mutant tumours who have already had cancer treatment.[21] The recommended phase 2 dose was established as 4·8 mg/kg once every 3 weeks.[21]

This study (the phase 2 portion of the phase 1/2 trial) aimed to further evaluate the anti-tumour activity and safety of trastuzumab rezetecan at the recommended phase 2 dose in patients with advanced *HER2*-mutant NSCLC who have received at least one previous line of systemic therapy.

Methods

Study design and participants

We conducted a multicentre, single-arm, phase 1/2 trial in 35 hospitals in China (appendix 2 pp 4, 14). The design and results of the dose-escalation and expansion phase 1 part of this trial have been published previously.[21] In this phase 2 part, the activity and safety of trastuzumab rezetecan at the recommended dose of 4·8 mg/kg was further assessed in patients with advanced *HER2*-mutant NSCLC.

Eligible patients were aged 18–75 years, with documented locally advanced or metastatic NSCLC harbouring an activating *HER2* mutation, as determined by a central laboratory in a tumour tissue sample using next-generation sequencing (Geneplus; Beijing, China). Patients must have previously received platinum-based chemotherapy and anti-PD-1 therapy or anti-PD-L1 therapy (either concurrently or sequentially) in the locally advanced or metastatic setting, and progressed on or after treatment, or been intolerant to these therapies. Additional inclusion criteria were an Eastern Cooperative Oncology Group (ECOG) performance status of 0–1, a life expectancy of at least 12 weeks, at least one measurable extracranial lesion per Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1, provision of fresh or archived (within 12 months) tumour tissue for biomarker analysis, and adequate organ functions. Key exclusion criteria included untreated or active CNS metastases (patients who had received adequate local treatment for CNS metastases ≥2 weeks before the first study dose and did not require corticosteroids were eligible); previous treatment with anti-HER2 agents (except for pan-HER tyrosine kinase inhibitors) or topoisomerase I-based antibody–drug conjugates; systemic anti-cancer treatment within 4 weeks (or within 5 half-lives for small-molecule targeted therapies); concurrent oncogenic driver mutations with approved targeted therapies; secondary *HER2* mutation after acquired resistance to EGFR-tyrosine kinase inhibitors; a history of immunodeficiency; uncontrolled or clinically significant cardiovascular disease; serious infections within the preceding 30 days; and a history of non-infectious interstitial lung disease that required steroids or current or suspected interstitial lung disease. The complete eligibility criteria are provided in the protocol (appendix 2). Patient sex and race were recorded at the time of enrolment per each participant's Resident Identity Card of China (no specific options were provided).

The study protocol and amendments were approved by the ethics committees of the lead clinical site (Shanghai Chest Hospital, Shanghai, China, #LS2115) and all participating sites. The study was conducted in accordance with the International Council for Harmonisation Good Clinical Practice guidelines, the Declaration of Helsinki, and applicable local regulations. All patients provided written informed consent before enrolment. Patient representatives were not involved in the study design, conduct, analysis, interpretation, or reporting. The study was registered with ClinicalTrials.gov, NCT04818333.

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See **Online** for appendix 2

Procedures

Enrolled patients received trastuzumab rezetecan at the recommended phase 2 dose of 4·8 mg/kg by intravenous infusion once every 3 weeks. Treatment continued until disease progression per RECIST version 1.1 (as assessed by an independent review committee), unacceptable toxicities, withdrawal of consent, initiation of new anti-cancer treatment, or investigator decision. Treatment beyond the first radiographic progression was allowed, if the investigator determined that the patient was still deriving clinical benefit. One dose level reduction (to 3·2 mg/kg once every 3 weeks) and dose interruptions (up to 28 days) were permitted for managing adverse events. Details on management of pulmonary toxicities, including interstitial lung disease, are available in the study protocol (appendix 2).

Tumour assessments by contrast-enhanced CT or MRI were done at baseline, every 6 weeks until 54 weeks, and then every 12 weeks thereafter until radiographic progression, start of new anti-tumour therapy, loss of follow-up, or death. Contrast-enhanced CT was the recommended imaging modality for the chest, abdomen, and pelvis; MRI was the preferred option for brain evaluation. Brain scans were mandatory at screening. After screening, brain imaging followed the same schedule as the thoracic or abdomen scans in patients with baseline brain metastases; for patients without baseline brain metastases, brain imaging was performed as clinically indicated for suspected new lesions. Tumour response was evaluated by the independent review committee and investigator according to RECIST version 1.1; all complete responses or partial responses required confirmation at least 4 weeks after the initial documentation of response. Safety was monitored until 40 days after administration of the last study dose, with adverse events graded by the investigator according to National Cancer Institute Common Terminology Criteria for Adverse Events version 5.0. After the end of the safety evaluation period, patients were followed up for survival every 60 days until withdrawal of consent, loss to follow-up, death, or study termination. Exploratory biomarkers were centrally analysed in formalin-fixed paraffin-embedded tissue samples, with *HER2* mutation and gene amplification (defined as a copy number ≥4) assessed by next-generation

sequencing (Geneplus; Beijing, China), and HER2 expression assessed by immunohistochemistry (OrigiMed; Shanghai, China). HER2 immunohistochemistry staining was conducted using the Roche Ventana anti-HER2/neu (4B5) antibody on the Ventana BenchMark ULTRA automated staining platform (Ventana Medical Systems; Tucson, AZ, USA). Immunohistochemistry scoring followed the American Society of Clinical Oncology/College of American Pathologists criteria developed for breast cancer (appendix 2 p 2).[22]

Outcomes

The primary endpoint was objective response rate, defined as the proportion of patients with a best overall response of complete response or partial response, as assessed by the independent review committee according to RECIST version 1.1. Secondary endpoints were objective response rate assessed by investigator; disease control rate (ie, the proportion of patients with a best overall response of complete response, partial response, or stable disease), duration of response (ie, the time from the first documented response to the first documented disease progression defined by RECIST version 1.1 or death from any cause), and progression-free survival (ie, the time from the first dose to the first documented disease progression defined by RECIST version 1.1 or death from any cause, whichever occurred first), all assessed by the independent review committee and investigator; overall survival (ie, the time from the first dose to death from any cause); and safety (incidence and severity of adverse events according to the National Cancer Institute Common Terminology Criteria for Adverse Events, version 5.0). Exploratory endpoints, including the pharmacokinetics and immunogenicity of trastuzumab rezetecan, will be reported in a separate publication.

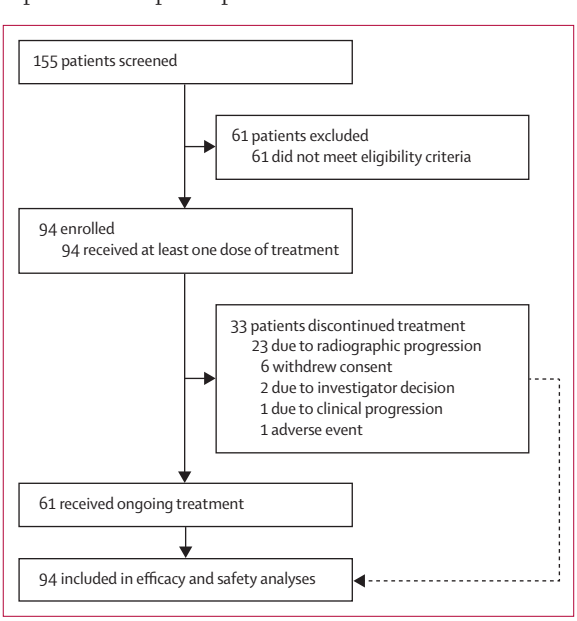


Figure 1: **Trial profile**

Statistical analysis

The phase 2 portion of the study planned to enrol 90 patients. This sample size was calculated to ensure approximately 83% probability that the lower limit of the 95% CI for objective response rate would exceed 25%, assuming an expected objective response rate of 40% for trastuzumab rezetecan. The benchmark objective response rate of 25% was the estimated upper limit for the historical standard-of-care, docetaxel, which had reported objective response rates of 5–15% in patients with previously treated NSCLC.[23–25] The primary endpoint of independent review committee-assessed objective response rate was met, if the lower limit of the 95% CI exceeded 25%.

Activity and safety were analysed in all patients who received at least one dose of study medication. All treated patients were included in the denominator for objective response rate and disease control rate analysis, with those having non-evaluable responses (no measurable baseline disease per RECIST version 1.1 or post-baseline assessment) classified as non-responders. The 95% CIs for objective response rate and disease control rate were calculated with the Clopper Pearson method. Prespecified (by sex, age, ECOG performance status, brain metastases at baseline, previous lines of treatment, and previous anti-HER2 treatment) and post-hoc (by smoking status) subgroup analyses and prespecified exploratory biomarker analysis (according to *HER2* mutation type and HER2 expression) were done for objective response rate per independent review committee. Median progression-free survival, duration of response, and overall survival were analysed using the Kaplan–Meier method, with the 95% CIs estimated using the Brookmeyer–Crowley methods with log-log transformation. Patients without an event before the data cutoff date were censored at their last valid assessment (appendix 2 pp 2–3). Survival and response rates were analysed post hoc at 6 months using the Kaplan–Meier method, with 95% CIs calculated via the pointwise method based on log–log transformation. Post-hoc Kaplan–Meier analysis of independent review committee-assessed progression-free survival by brain metastasis at baseline and *HER2* mutation (ie, Ala775_Gly776insTyrValMetAla *vs* other mutations) was also conducted. Safety data were summarised using descriptive statistics. All analyses were done with SAS version 9.4. There was no data monitoring committee involved in the trial.

Role of the funding source

The funder collaborated with academic authors regarding the study design, data collection, data analysis, data interpretation, and writing of the report.

Results

Between April 14, 2023, and Dec 14, 2023, 155 patients were screened. 61 patients who did not meet eligibility criteria were excluded at screening; 94 patients were enrolled. All 94 patients received at least one dose of

trastuzumab rezetecan and were included in the efficacy and safety analyses (figure 1). As of the data cutoff date on June 14, 2024, the median duration of follow-up was 8·7 months (IQR 7·0–10·4). 61 (65%) of 94 patients remained on treatment; the most common reason for treatment discontinuation was radiographic progression, which occurred in 23 (24%) patients. 16 (17%) patients received post-study systemic anticancer therapy (appendix 2 p 5).

Baseline demographic and clinical characteristics of patients are presented in table 1. 42 (45%) patients were male, 52 (55%) female, 92 (98%) were Han Chinese, and two (2%) were other ethnicity Chinese. The median number of previous therapies was 2 (IQR 1–2); all 94 patients had received platinum-based chemotherapy and anti-PD-1 therapy or anti-PD-L1 therapy, with 22 (23%) previously treated with an anti-HER2 tyrosine kinase inhibitor. 24 (26%) patients had brain metastasis at study entry. The most common *HER2* mutations were exon 20 insertions, particularly Ala775_Gly776insTyrValMetAla, which was present in 68 (72%) patients.

At data cutoff, 69 (73%; 95% CI 63·3–82·0) of 94 patients had a confirmed objective response (all partial responses), as assessed by independent review committee (figure 2A, 2C; appendix 2 p 6).

An additional 23 (25%) patients had stable disease, with most experiencing a reduction in the size of target lesions (figure 2A). The disease control rate was 98·9% (95% CI 94·2–100·0; appendix 2 p 6). In patients with an objective response, the median duration of response was not reached (95% CI 8·3–not reached); the ongoing response rate at 6 months (post-hoc analysis) was 83·9% (70·0–91·8); and 51 (74%) of 69 patients had an ongoing response at data cutoff (figure 2B, figure 3A; appendix 2 p 7). Responses with trastuzumab rezetecan were consistently observed across prespecified and post-hoc baseline subgroups, irrespective of previous lines of therapy, previous anti-HER2 tyrosine kinase inhibitor use, or presence of brain metastasis (figure 2C). In the pre-specified exploratory biomarker analyses, anti-tumour activity was also evident across various activating *HER2* mutations, including single-nucleotide variants and exon 20 insertions, and a range of HER2 expression levels from immunohistochemistry scores 0 to 2+ (appendix 2 p 8). Of the two patients with squamous cell histology, one had a partial response and the other had stable disease (appendix 2 p 9). Tumour response based on investigator assessment was consistent with independent review committee findings (appendix 2 p 6).

As of data cutoff, 25 (27%) of 94 patients had disease progression per independent review committee, or died. The median progression-free survival was 11·5 months (95% CI 9·9–not reached; figure 3B); the progression-free survival rate at 6 months (post-hoc analysis) was 86·8% (77·4–92·5). Censoring reasons are provided in appendix 2 (p 7). In post-hoc analysis, the median progression-free survival was 9·9 months (8·4–not reached) in patients with brain metastasis at baseline and 11·5 months (10·5–not reached) in patients without brain metastasis. By *HER2* mutation, the median

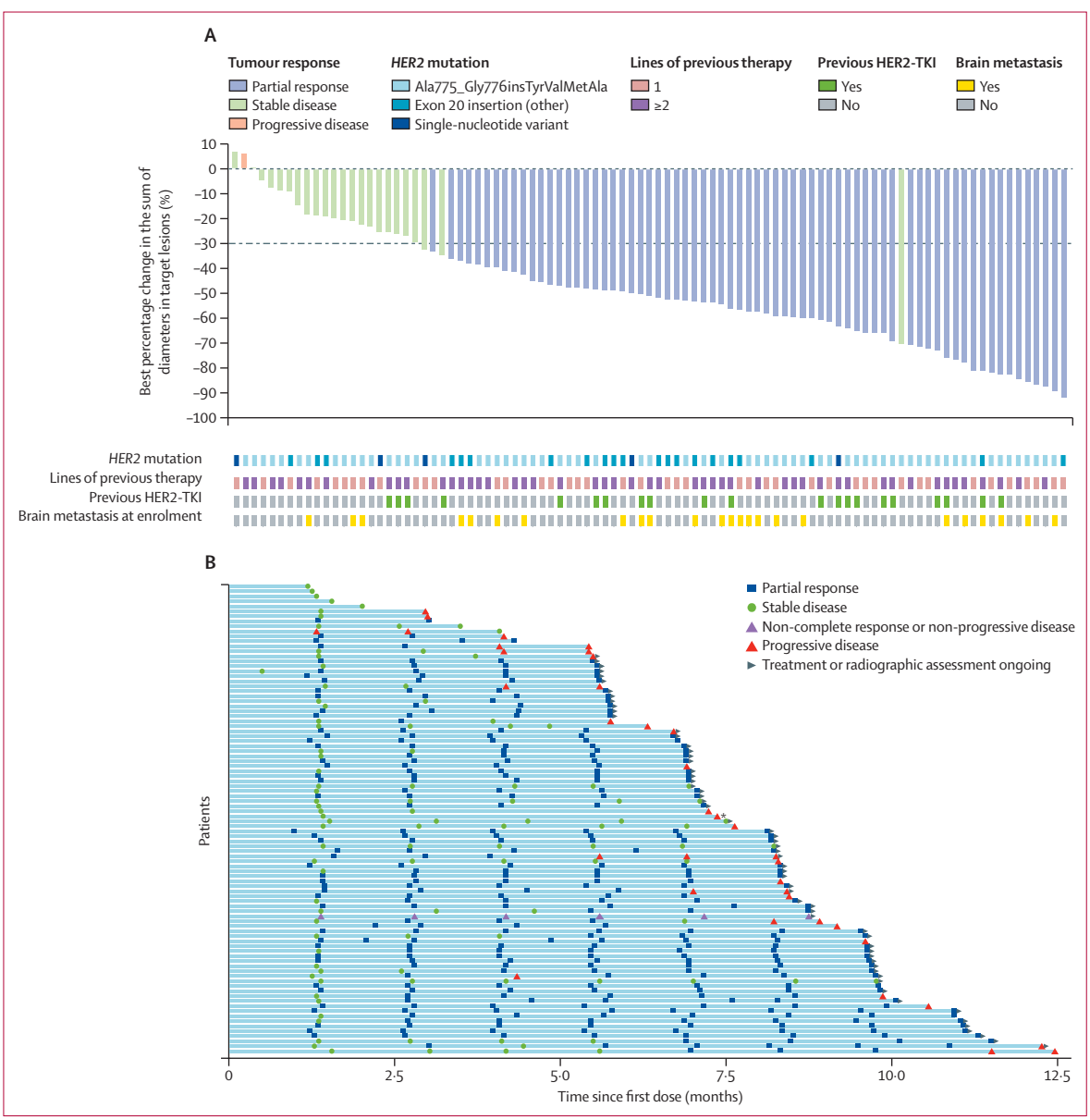
	Patients
Median age, years	57·5 (50–64)
Sex	
Male	42/94 (45%)
Female	52/94 (55%)
Ethnicity	
Han Chinese	92/94 (98%)
Other ethnicity Chinese	2/94 (2%)
ECOG performance status	
0	18/94 (19%)
1	76/94 (81%)
Smoking status	
Current or former	34/94 (36%)
Never	60/94 (64%)
Histology	
Adenocarcinoma	91/94 (97%)
Squamous	2/94 (2%)
Other	1/94 (1%)
TNM clinical stage	
TNM stage IV	92/94 (98%)
TNM stage IIIB	2/94 (2%)
Site of metastases at baseline	
Brain	24/94 (26%)
Liver	24/94 (26%)
Lines of previous therapy	2 (1–2)
Type of previous therapy	
Platinum-based chemotherapy	94/94 (100%)
Anti-PD-1 treatment or anti-PD-L1 treatment	94/94 (100%)
HER2-tyrosine kinase inhibitor	22/94 (23%)
Location of *HER2* mutations	
Kinase domain	91/94 (97%)
Extracellular domain	3/94 (3%)
HER2 mutations*	
Ala775_Gly776insTyrValMetAla	68/94 (72%)
Gly776>ValCys	11/94 (12%)
Pro780_Tyr781insGlySerPro	6/94 (6%)
Gly776>LeuCys	4/94 (4%)
Other†	5/94 (5%)
HER2 expression*‡	
Immunohistochemistry 0	40/81 (49%)
Immunohistochemistry 1+	24/81 (30%)
Immunohistochemistry 2+	17/81 (21%)
HER2 amplification*	
Yes	1/94 (1%)
No	93/94 (99%)

Data are median (IQR) or n/N (%). ECOG=Eastern Cooperative Oncology Group. *Central tissue-based testing. †Ser310Phe (n=3), Leu755Pro (n=1), and Leu869Arg/Thr862Ala (n=1). ‡Data available in 81 patients, HER2 expression was not evaluated in the remaining patients due to insufficient tumour tissue availability.

Table 1: **Baseline characteristics**

progression-free survival was 12·3 months (9·9–not reached) in patients with the Ala775_Gly776insTyrValMetAla mutation and 11·5 months (7·6–not reached) in patients with other mutations. Results of investigator-assessed progression-free survival are provided in appendix 2 (p 6). Overall survival data were not mature, with three (3%) deaths in 94 patients (appendix 2 p 15).

The median treatment duration with trastuzumab rezetecan was 8·0 months (IQR 6·4–10·1). All 94 (100%) patients had at least one treatment-related adverse event, with grade 3 or worse events reported in 58 (62%) patients (table 2). The most common grade 3–4 treatment-related adverse events were decreased neutrophil count (38 [40%] patients), decreased white blood cell count (25 [27%]), anaemia (22 [23%]), decreased platelet count (10 [11%]), and decreased lymphocyte count (seven [7%]). Interstitial lung disease was reported in seven (7%) patients, with a median time to onset of 84 days (IQR 78–248; table 2). Treatment-related adverse events led to dose interruption in 28 (30%) of 94 patients, dose reduction in 16 (17%) patients, and dose discontinuation in one (1%) patient (grade 3 interstitial lung disease). Serious treatment-related adverse events were reported in 22 (23%) of 94 patients, with the most common being decreased platelet count (six [6%]), decreased neutrophil count (six [6%]), interstitial lung disease (five [5%]), decreased white blood cell count (four [4%]), anaemia (four [4%]),



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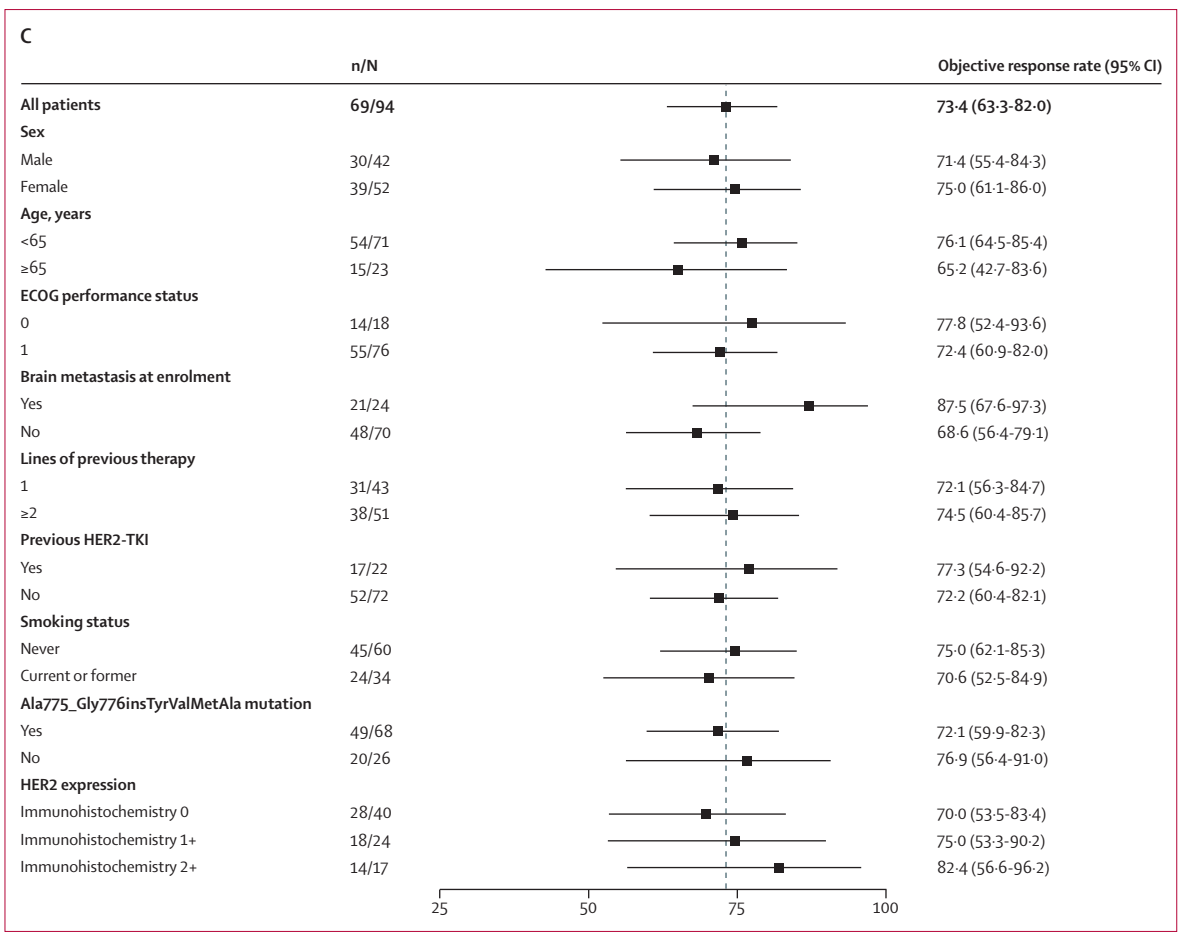

Figure 2: **Tumour response per independent review committee**
(A) Waterfall plot of best percentage change of tumour size from baseline to data cutoff (ie, June 14, 2024). Dashed lines indicate 30% decrease in tumour size. Data are presented for 93 patients; one patient with no measurable disease (assessed as having non-complete response and non-progressive disease) is not included. (B) Response and duration of treatment of trastuzumab rezetecan in all 94 patients who received treatment. (C) Objective response rate across patient subgroups at data cutoff (ie, June 14, 2024). n/N are number of responders out of number of patients. All subgroups, except those defined by smoking status, were prespecified. TKI=tyrosine kinase inhibitor. *This patient missed multiple scheduled tumour assessments following the first post-baseline assessment, due to cholelithiasis and infective cholecystitis (treatment unrelated, as determined by the investigator). ECOG=Eastern Cooperative Oncology Group.

vomiting (three [3%]), pneumonia (three [3%]), hyponatraemia (two [2%]), pyrexia (one [1%]), small intestinal obstruction (one [1%]), nausea (one [1%]), and chronic obstructive pulmonary disease (one [1%]). Lists of common adverse events, adverse events leading to dose modifications, and serious adverse events, regardless of attribution, are available in appendix 2 (pp 10–13). Two patients died due to adverse events (progression of malignant neoplasm), both assessed as unrelated to study treatment by the investigator.

Discussion

The phase 2 part of this trial, evaluating the activity and safety of trastuzumab rezetecan at the recommended phase 2 dose in pretreated patients with *HER2*-mutant NSCLC, confirmed and expanded on the findings from the phase 1 part of the study.[21] The primary endpoint was achieved, with trastuzumab rezetecan showing an objective response rate of 73% (95% CI 63·3–82·0), as assessed by an independent review committee. The anti-tumour activity was consistent across clinically relevant subgroups, regardless of previous lines of therapy, previous anti-HER2 tyrosine kinase inhibitor treatment, or the presence of brain metastases at baseline. Responses with trastuzumab rezetecan were durable, with more than two-thirds of responses ongoing at data cutoff, leading to a clinically meaningful progression-free survival. The safety profile of trastuzumab rezetecan was manageable, with a low rate of treatment discontinuation and no deaths due to treatment-related adverse events.

HER2 mutations are well established oncogenic drivers in NSCLC and have been extensively explored as a therapeutic target.[3–9,13] However, antibody-based therapies and tyrosine kinase inhibitors have yielded mixed results, and no HER2-directed treatments were approved for NSCLC until trastuzumab deruxtecan. In the

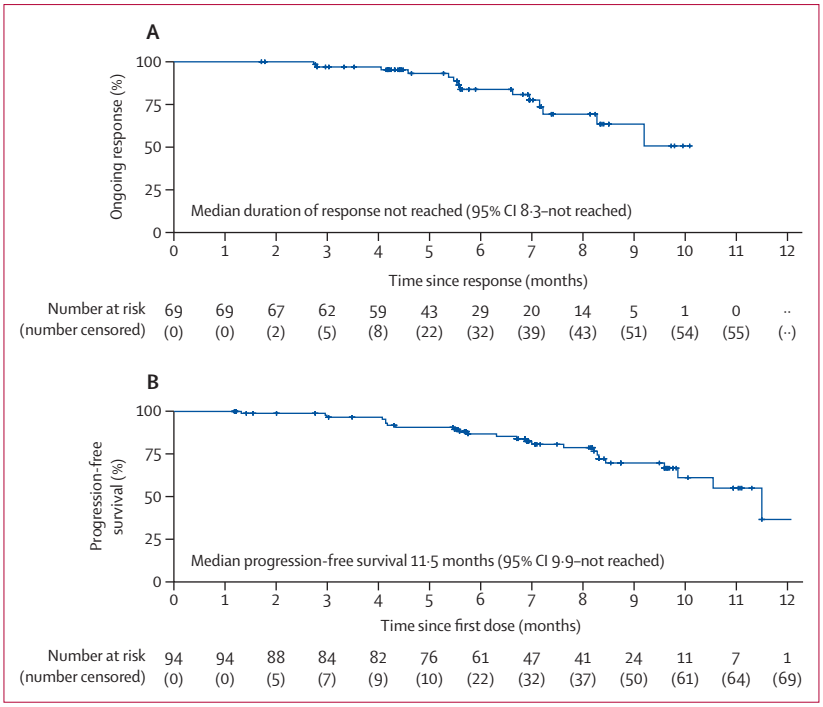

Figure 3: **Kaplan–Meier analysis of duration of response and progression-free survival as assessed by the independent review committee**
(A) Duration of response in responders. (B) Progression-free survival. Data cutoff date was June 14, 2024.

	Grades 1–2	Grade 3	Grade 4
Haematological toxicities			
Neutrophil count decreased	44 (47%)	29 (31%)	9 (10%)
White blood cell count decreased	52 (55%)	24 (25%)	1 (1%)
Anaemia	53 (56%)	22 (23%)	0
Platelet count decreased	26 (28%)	8 (9%)	2 (2%)
Lymphocyte count decreased	5 (5%)	6 (6%)	1 (1%)
Non-haematological toxicities			
Nausea	50 (53%)	0	0
Decreased appetite	45 (48%)	0	0
Asthenia	39 (41%)	1 (1%)	0
Vomiting	35 (37%)	2 (2%)	0
Alopecia	33 (35%)	0	0
Hypertriglyceridaemia	32 (34%)	1 (1%)	0
Aspartate aminotransferase increased	29 (31%)	0	0
Alanine aminotransferase increased	25 (27%)	2 (2%)	0
Hypoalbuminaemia	26 (28%)	0	0
γ-glutamyltransferase increased	20 (21%)	2 (2%)	0
Weight decreased	18 (19%)	0	0
Hypercholesterolaemia	18 (19%)	0	0
Weight increased	15 (16%)	0	0
Stomatitis	14 (15%)	1 (1%)	0
Blood lactate dehydrogenase increased	14 (15%)	0	0
Hyperglycaemia	13 (14%)	0	0
Constipation	13 (14%)	0	0
Urinary tract infection	12 (13%)	0	0
Blood creatinine increased	11 (12%)	0	0
Hypocalcaemia	10 (11%)	1 (1%)	0
Hyponatraemia	9 (10%)	1 (1%)	1 (1%)
Diarrhoea	10 (11%)	1 (1%)	0
Blood alkaline phosphatase increased	10 (11%)	0	0
Proteinuria	10 (11%)	0	0
Hypokalaemia	7 (7%)	3 (3%)	0
Pneumonia	6 (6%)	3 (3%)	0
Interstitial lung disease	6 (6%)	1 (1%)	0
Fatigue	4 (4%)	1 (1%)	0
Ejection fraction decreased	2 (2%)	1 (1%)	0
Vulvitis	0	1 (1%)	0
Small intestinal obstruction	0	1 (1%)	0

Data are number of patients (%). Treatment-related adverse events of grade 1–2 occurring in more than 10% of patients, and all grade 3 or higher events are listed. No grade 5 events were reported.

Table 2: **Treatment-related adverse events**

DESTINY-Lung01 trial, trastuzumab deruxtecan at 6·4 mg/kg resulted in an objective response in 50 (55%) of 91 patients (95% CI 44–65), a median duration of response of 9·3 months (5·7–14·7), and a median progression-free survival of 8·2 months (6·0–11·9) in pretreated patients with *HER2*-mutant NSCLC.[16] In the subsequent dose-optimisation DESTINY-Lung02 trial, trastuzumab deruxtecan at 5·4 mg/kg showed reduced toxicity and preserved anti-tumour activity, with an objective response rate of 50·0% (95% CI 39·9–60·1), a median duration of response of 12·6 months (6·4–not reached), and a median progression-free survival of 10·0 months (7·7–15·2),[18] leading to the approval of the lower dose. Similar activity was observed for trastuzumab deruxtecan in Chinese patients with *HER2*-mutant NSCLC in the DESTINY-Lung05 trial.[26] In 2024, zongertinib and BAY 2927088 emerged as promising new anti-HER2 tyrosine kinase inhibitors for *HER2*-mutant NSCLC in early-phase trials,[10,11] achieving notable objective response rates (66·7–78·2% for zongertinib and 72·1% for BAY 2927088) and durable responses (median duration of response not reached for zongertinib and 8·7 months for BAY 2927088). Tyrosine kinase inhibitors can be administered orally and have a distinct safety profile to antibody-based therapies;[10,11] further data from extended follow-up and larger scale studies are eagerly anticipated. Although cross-trial comparisons should be made with caution, the clinical benefits with trastuzumab rezetecan appeared favourable in the context of currently available therapies. The robust activity of trastuzumab rezetecan, despite a lower drug–antibody ratio than trastuzumab deruxtecan, was supported by its specific payloads, which showed superior membrane permeability and enhanced cytotoxic activity compared with that of trastuzumab deruxtecan in the preclinical study.[19] Similar to trastuzumab deruxtecan, clinical activity with trastuzumab rezetecan persisted in

patients with brain metastases—a prevalent, but historically challenging-to-treat, subgroup,[27,28] highlighting the broad spectrum of patients who might benefit. In line with previous findings, trastuzumab rezetecan showed activity across various *HER2* mutations (single-nucleotide variants and exon 20 insertions), even in the absence of detectable HER2 expression, potentially due to increased ubiquitination and internalisation of the antibody–drug conjugate–receptor complex in the presence of activating *HER2* mutations.[29] Additionally, trastuzumab rezetecan remained effective in patients with previous anti-HER2 tyrosine kinase inhibitor use, indicating low potential for cross-resistance due to different mechanism of action.

Although both the phase 1 and 2 portions of this study showed substantial and durable activity of trastuzumab rezetecan in *HER2*-mutant NSCLC, differences in objective response rate and progression-free survival were noted between phase 1 and phase 2 (ie, phase 1 objective response rate 41·9% and median progression-free survival 8·4 months per investigator).[21] The baseline patient characteristics were largely similar, with the phase 2 portion including a lower proportion of patients with an ECOG performance status of 1 and previous anti-HER2 tyrosine kinase inhibitor use, but a higher proportion with previous immune checkpoint inhibitor treatment. However, none of these factors appeared to fully account for the variances in objective response rate based on subgroup analysis. Alternatively, the outcomes in the phase 1 portion might have been negatively affected by the COVID-19 pandemic[30] and a higher rate of treatment interruptions, whereas the phase 2 results could have benefited from accumulated experience in managing trastuzumab rezetecan-related adverse events, which facilitated treatment administration.

The safety profile of trastuzumab rezetecan observed in this study was generally consistent with that of trastuzumab deruxtecan[16,17] and aligned with results from phase 1 trials of trastuzumab rezetecan,[20,21] with no new safety signals identified. The rate of treatment discontinuation due to treatment-related adverse events was low (1%), in line with previous studies of trastuzumab rezetecan (2·6–4·7%).[20,21] This low rate is attributed to the design features of trastuzumab rezetecan, including a moderate drug–antibody ratio and a chiral cyclopropyl group between the linker and toxin, which improves stability and minimises toxicities from premature toxin release. The most common grade 3 or worse treatment-related adverse events were haematological adverse events, potentially attributable, at least in part, to the strong bystander effect from toxins released by target cancer cells. The incidence of haematological adverse events was largely similar to that reported for trastuzumab deruxtecan in the DESTINY-Lung05 trial in Chinese patients, but higher than in the global DESTINY-Lung01/02 trials.[16,17,26] A global phase 1 trial of trastuzumab rezetecan in advanced solid tumours showed a similar trend, with Asian individuals having a higher risk of haematological adverse events than non-Asian individuals.[20] Haematological events could be efficiently managed with standard supportive care without compromising treatment administration. Interstitial lung disease is a notable risk associated with trastuzumab deruxtecan, occurring in 9·7–12·9% (grade ≥3 in 1·4–2·0%) of patients with advanced NSCLC at the approved dose.[17,26] In this study, interstitial lung disease occurred in seven (7%) patients after a similar median duration of exposure. Most cases were mild, with one (1%) grade 3 event, indicating a relatively low risk of interstitial lung disease. Active monitoring, timely dose interruption, and corticosteroid intervention per protocol partly contributed to the low incidence of high-grade events.

Limitations of this study include the absence of a control group and the moderate sample size for subgroup analysis. Currently, there are no plans for a randomised trial for trastuzumab rezetecan for advanced *HER2*-mutant NSCLC in a second-line or later setting, given the low activity of available conventional therapies. However, a randomised, phase 3 trial (NCT06430437) is underway to evaluate trastuzumab rezetecan versus platinum-based chemotherapy plus immunotherapy as a first-line treatment for patients with *HER2*-mutant NSCLC. Another limitation was that patients with brain metastases at baseline were required to have received adequate local treatment before enrolment. Therefore, most patients had no measurable CNS disease at study entry, limiting the evaluation of intracranial response. Further studies are warranted to evaluate the CNS activity of trastuzumab rezetecan. Furthermore, overall survival data are not yet mature. Follow-up is ongoing to provide a more comprehensive understanding of long-term survival and safety. Lastly, only Chinese patients were included. The activity and safety of trastuzumab rezetecan in patients with NSCLC of other ethnic groups remains to be established, although preliminary data from a global phase 1 trial indicate similar pharmacokinetics and anti-tumour activity of trastuzumab rezetecan across Asian and non-Asian patients in a pan-tumour setting.[20]

In summary, trastuzumab rezetecan showed clinically meaningful activity and a manageable safety profile in patients with previously treated *HER2*-mutant NSCLC, providing a potential new treatment option for this population. The substantial and durable activity observed in this study supports further investigation of trastuzumab rezetecan in earlier disease stages.

Contributors

SL and WS were responsible for study design and supervision. SL, ZL, YaW, YSu, LW, XL, LS, ZH, HY, YongsW, QW, ZS, WH, YongW, GX, YY, MP, YSo, DW, RM, JF, YLu, WL, SH, and ZW were involved in patient enrolment and data collection. KS conducted the statistical analysis. All authors critically revised the manuscript for intellectual content and contributed to data interpretation. ZL, KS, YLi, and LY verified study

data. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Declaration of interests

SL reports research grant from AstraZeneca, Abbvie, Roche, Bristol-Myers Squibb, Hutchison Whampoa, BeiGene, Daiichi Sankyo, Bayer, FibroGen, Hansoh Pharma, Hengrui, Innovent, Zai Lab, Betta, Akesobio, and Inventisbio. KS, YLi, LY, and WS are employees of Jiangsu Hengrui Pharmaceuticals. All other authors declare no competing interests.

Data sharing

Individual deidentified participant data that underlie the results reported in this Article can be requested from 24 months after study completion for an indefinite time period. Researchers should submit a proposal to the corresponding author outlining the reasons for requiring the data. The leading clinical site and sponsor will check whether the request is subject to any intellectual property or confidentiality obligations. A signed data access agreement with the sponsor is required before accessing shared data. The study protocol is provided in appendix 2.

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